

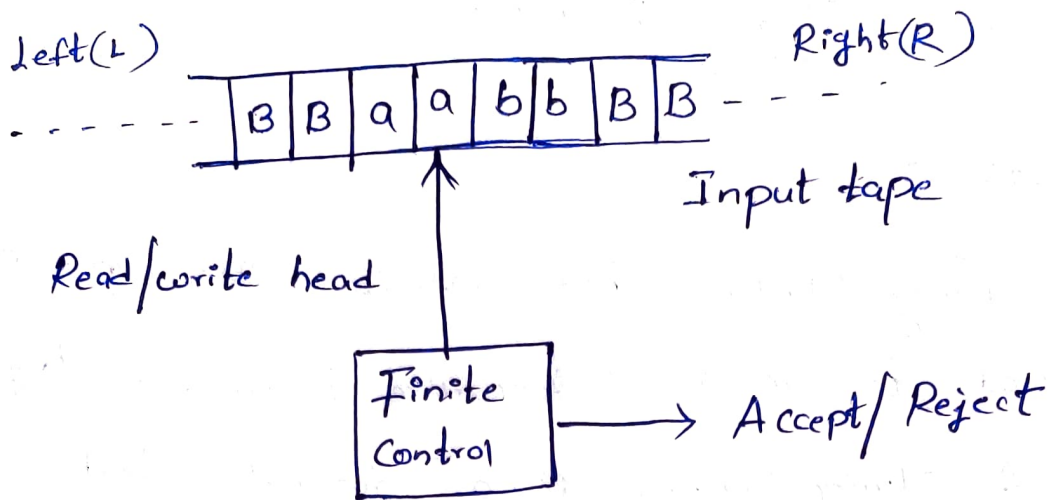
## Unit-V

# Turing Machines

### Introduction :

- Turing Machine has infinite size tape and it is used to accept "Recursive Enumerable language".
- A Turing Machine is a automata that can read, write and erase symbols in an infinite long tape.
- Turing Machine is most powerful than all remaining automata like Finite Automata, Push Down Automata and Linear bounded automata.
- It is called as Type-0 (zero) grammar, i.e. it supports "Unrestricted grammar" and "Recursively Enumerable language".
- Turing Machines are the fundamental concept in the theory of computation and play a vital role in Computer Science.
- It can produce a certain output based on its input.

## Block diagram (or) Model of Turing Machine:



- Turing Machine model mainly contains '3' components. They are

- i) Input Tape
- ii) Read/write head
- iii) Finite control

### Input Tape:

- It contains infinite length and contains infinite cells.

- Each cell either may contain input symbol or special symbol called blank (B).

- Read and write operations are performed on input tape.

### Read/write head :

- Read or write head reads or writes the symbols into the input tape.
- After reading an input symbol, it is replaced with another symbol, its state is changed and moves from one cell to another to "right" & "left" in input tape.

### Finite Control :

- Finite control is the one actually reads or writes input (or) symbols into input tape through the read/write head.
- It contains all the states of Turing machine and changes according to input.
- In Turing Machines, when reached to final state then input string is accepted (or) rejected. This is decided by the "Finite control".

## Formal Definition of Turing Machine :

Turing Machine mathematically represented by 7-tuple:

$$M = (Q, \Sigma, \Gamma, q_0, F, B, \delta)$$

$Q$  - Finite non empty set of states

$\Sigma$  - The set of non-empty input symbols called as "alphabet"

$\Gamma$  - The tape symbol ( $\emptyset$ ) Input tape alphabet

$q_0$  - Initial state

$F$  - Set of final states

$B$  - Blank symbol, used as end marker of input

$\delta$  - Transition function

### Transition function( $\delta$ ):

- Transition function( $\delta$ ) of Turing machine

is given as:

$$\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$



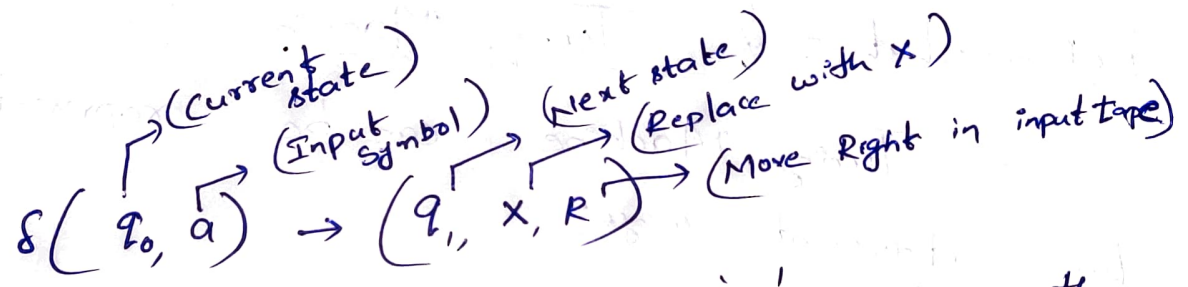
where  $Q$  - Finite non-empty set

$\Gamma$  - The ~~tape~~ input tape symbols

$L$  - Move read head towards "left"  
for reading symbol

$R$  - Move read head towards "right"  
for reading symbol.

Example:



- Here 'a' input symbol at state  $q_0$  gives the new state  $q_1$  and replacing symbol 'a' with 'x' on the input tape and move towards right (R) side.

$$\delta(q_2, b) \rightarrow (q_3, y, L)$$

- Here 'b' input symbol at state  $q_2$  gives the new state  $q_3$  and replaces symbol 'b' with 'y' on the input tape and move towards left side (L)

## operation of Turing Machine :

1. Read a symbol pointed by the read head
2. write a symbol pointed by read head
3. move read head either left or right sides on input tape.

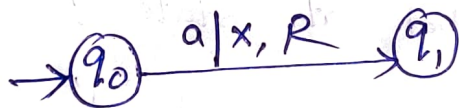
## Representation of Turing Machine :

- Turing Machine can be represented in two different ways :

1. Transition diagram
2. Transition table.

Ex:

$$f(q_0, a) \rightarrow (q_1, x, R)$$



### Transition diagram of Turing Machine

where  $q_0, q_1$  - states  
 $a, x$  - symbols in input tape  
 $R$  - Move right in input tape.

Example: Design Turing Machine for a language

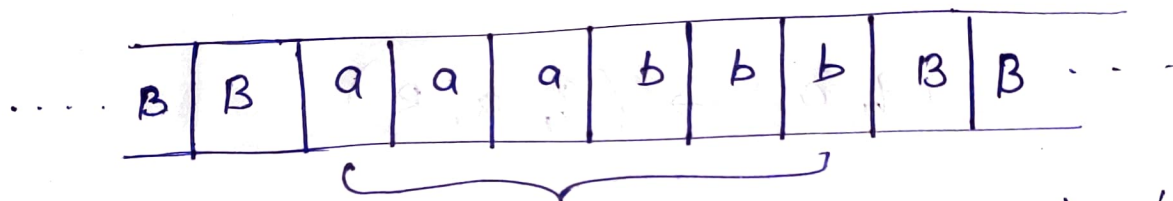
$$L = \{a^n b^n\} \text{ where } n \geq 1$$

Sol:

$$L = \{ab, aabb, aaabbb \dots\}$$

consider input tape with input string to be processed

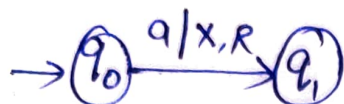
String: aaabbb



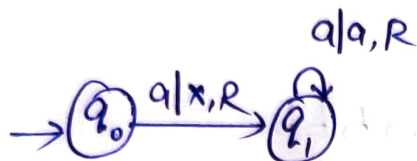
- Take first symbol of string at state  $q_0$  and replace 'a' with 'x'  $\rightarrow$  (By changing state to  $q_1$ ) and move towards right 'R' until we get another symbol 'b'
- Replace 'b' with 'y'  $\rightarrow$  (By changing the states) and move towards left 'L' until you reach 'x'
- Again move towards right (R) until we reach 'b' and replace with 'y' and move towards left to reach 'x'
- Repeat this steps until we reached to blank symbol (B)

↓  
B|B|a|a|a|b|b|b|B|B

now  $(q_0, a) \rightarrow q_1, \cancel{a/x} (q_1, x, R)$



↓  
B|B|x|a|a|b|b|b|B|B



↓  
B|B|x|a|a|b|b|~~B~~|B|B

$(q_1, a) \rightarrow (q_1, a, R)$

$(q_1, b) \rightarrow (q_2, y, L)$

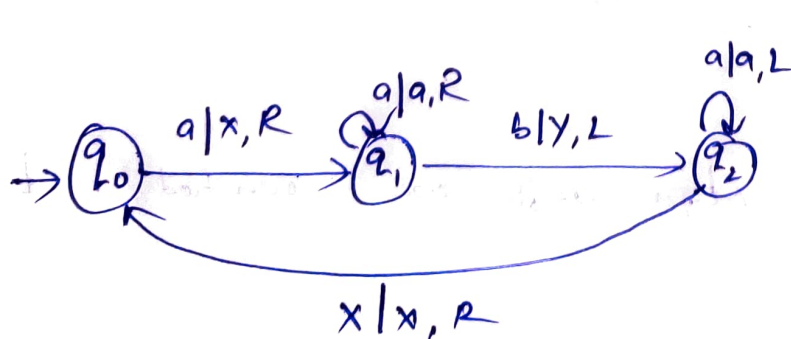
↓  
B|B|x|a|a|b|b|~~B~~|R|B



↓  
B|B|x|a|a|y|b|~~B~~|B|B



↓  
B|B|x|a|a|y|b|~~B~~|B|B



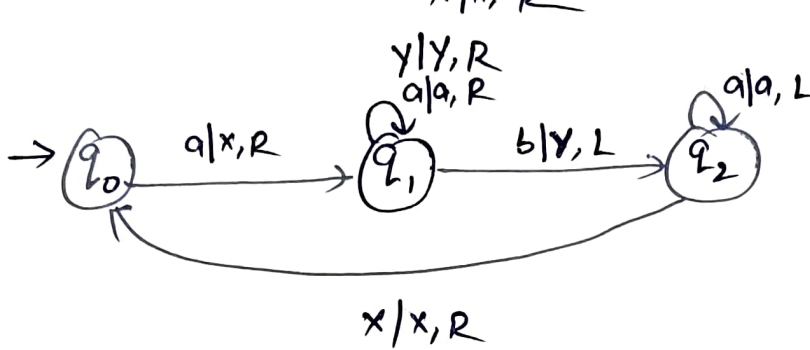
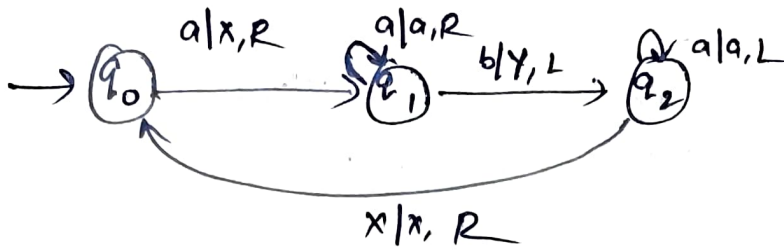
↓  
B|B|x|a|a|y|b|~~B~~|B|B

$(q_2, x) \rightarrow (q_0, x, R)$

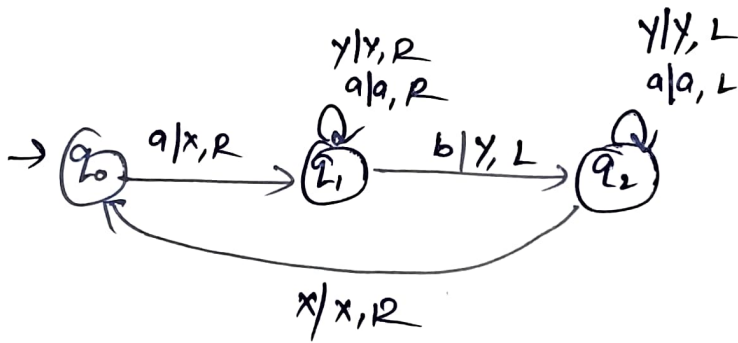


$$(q_0, a) \rightarrow (q_1, x, R)$$

$\downarrow$   
 $B|B|x|a|a|y|b|b|B|B$   
 $B|B|x|x|a|y|b|b|B|B$



$\downarrow$   
 $B|B|x|x|a|y|b|b|B|B$



$\downarrow$   
 $B|B|x|x|a|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|a|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|a|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|a|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|a|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

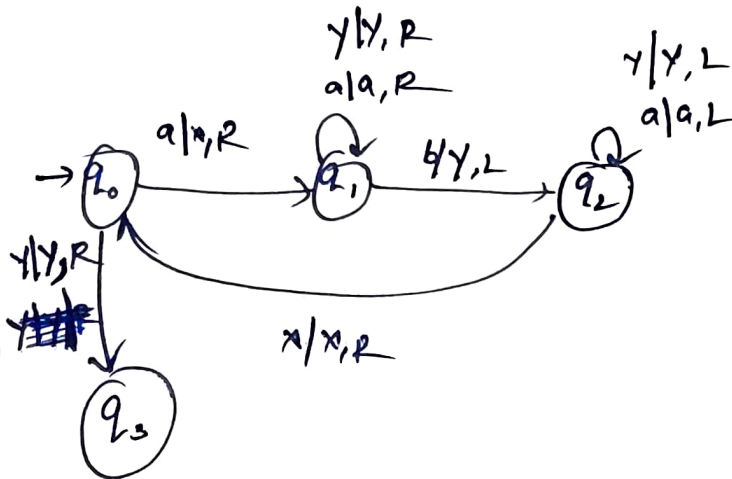
$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

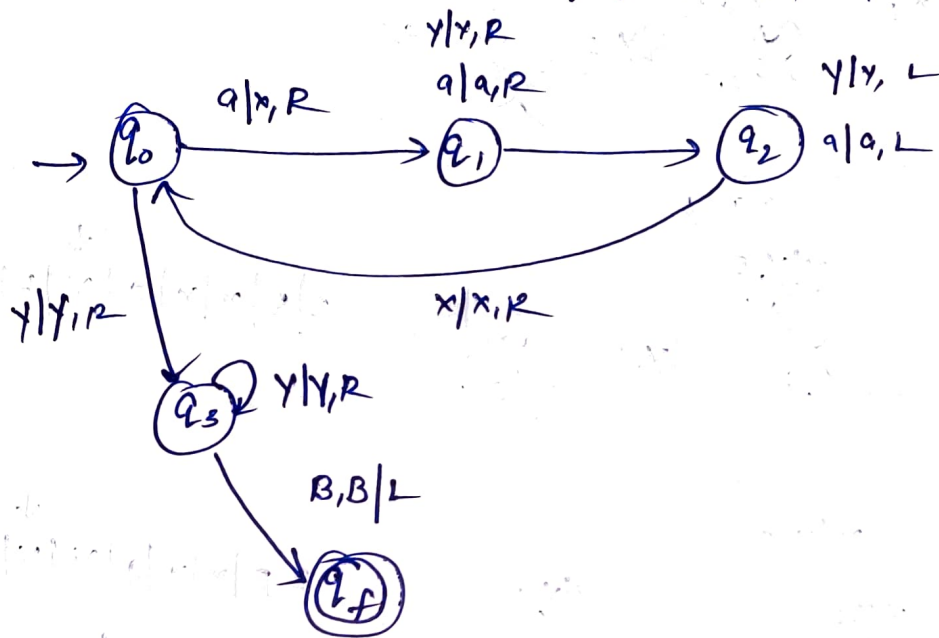
$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$

$\downarrow$   
 $B|B|x|x|x|y|y|b|b|B|B$



$B|B|x|x|x|y|y|y|B|B$

$B|B|x|x|x|y|y|y|B|B$

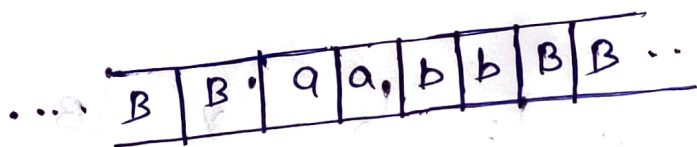


Example: Design Turing Machine for a Language and represent TM in transition diagram and transition table.

Sol: Given Language  $(L) = \{ab, aabb, aaabbb, \dots\}$

Now consider input tape with input string to be processed.

String : aabb



Transitions:

$$\delta(q_0, a) \rightarrow (q_1, x, R)$$

$$\delta(q_1, a) \rightarrow (q_1, a, R)$$

$$\delta(q_1, \gamma) \rightarrow (q_1, \gamma, R)$$

$$\delta(q_1, b) \rightarrow (q_2, \gamma, L)$$

$$\delta(q_2, a) \rightarrow (q_2, a, L)$$

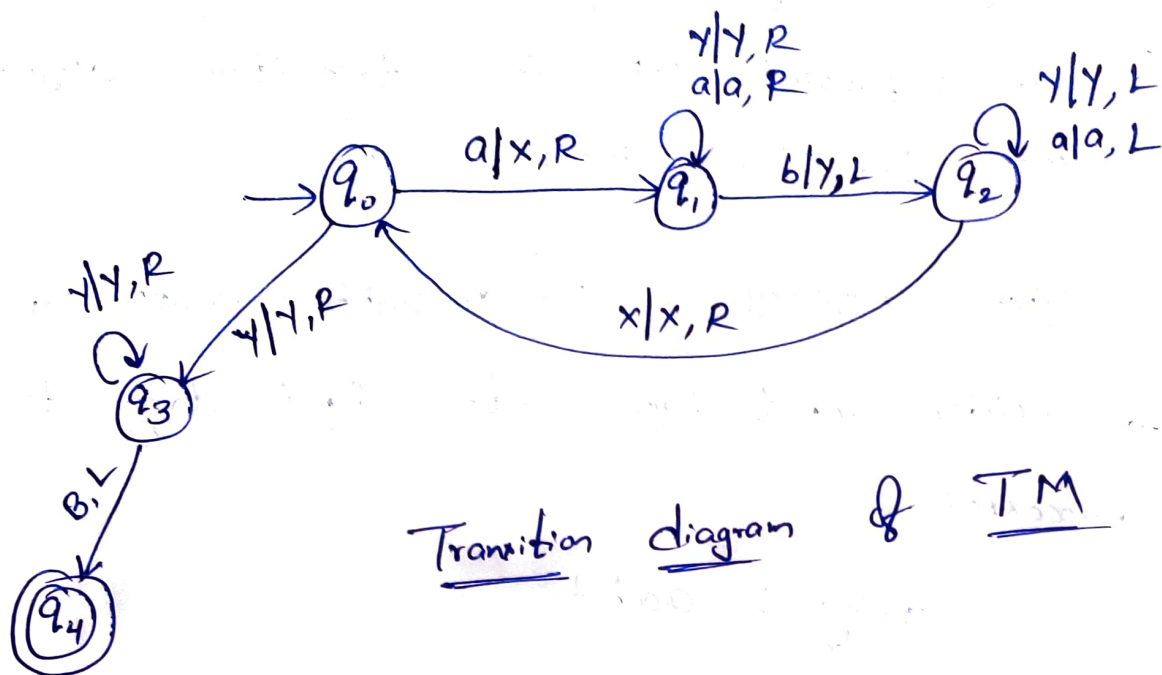
$$\delta(q_2, \gamma) \rightarrow (q_2, \gamma, L)$$

$$\delta(q_2, x) \rightarrow (q_0, x, R)$$

$$\delta(q_0, \gamma) \rightarrow (q_3, \gamma, R)$$

$$\delta(q_3, \gamma) \rightarrow (q_3, \gamma, R)$$

$$\delta(q_3, B) \rightarrow (q_4, B, L)$$



Transition diagram of TM

| States           | a                       | b                       | x                       | y                       | B                       |
|------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| → q <sub>0</sub> | (q <sub>1</sub> , x, R) |                         |                         | (q <sub>3</sub> , y, R) |                         |
| q <sub>1</sub>   | (q <sub>1</sub> , a, R) | (q <sub>2</sub> , y, L) |                         | (q <sub>1</sub> , y, R) |                         |
| q <sub>2</sub>   | (q <sub>2</sub> , a, L) |                         | (q <sub>0</sub> , x, R) | (q <sub>2</sub> , y, L) |                         |
| q <sub>3</sub>   |                         |                         |                         | (q <sub>3</sub> , y, R) | (q <sub>4</sub> , B, L) |
| q <sub>4</sub>   |                         |                         |                         |                         |                         |

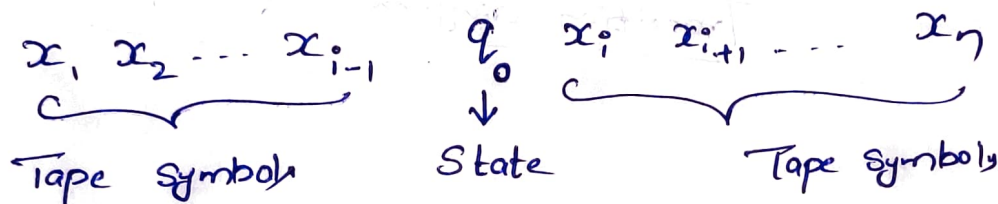
Transition table of TM



# Instantaneous Description of Turing Machine (TM):

Definition: Instantaneous Description (ID) of Turing machine is a snapshot of Turing machine to describe the current situation of the Turing machine.

Example: Let us consider the initial ID of a TM is:



On reading the symbol and move read-head to "left-side"

$$x_1, x_2, \dots, x_{i-1}, q_0, x_i, x_{i+1}, \dots, x_n \xrightarrow{\delta(q_0, x_i) \rightarrow (q_1, x, L)} x_1, x_2, \dots, x_{i-2}, q_1, x_{i-1}, x_i, x_{i+1}, \dots, x_n$$

On reading the symbol and move read-head to "right side"

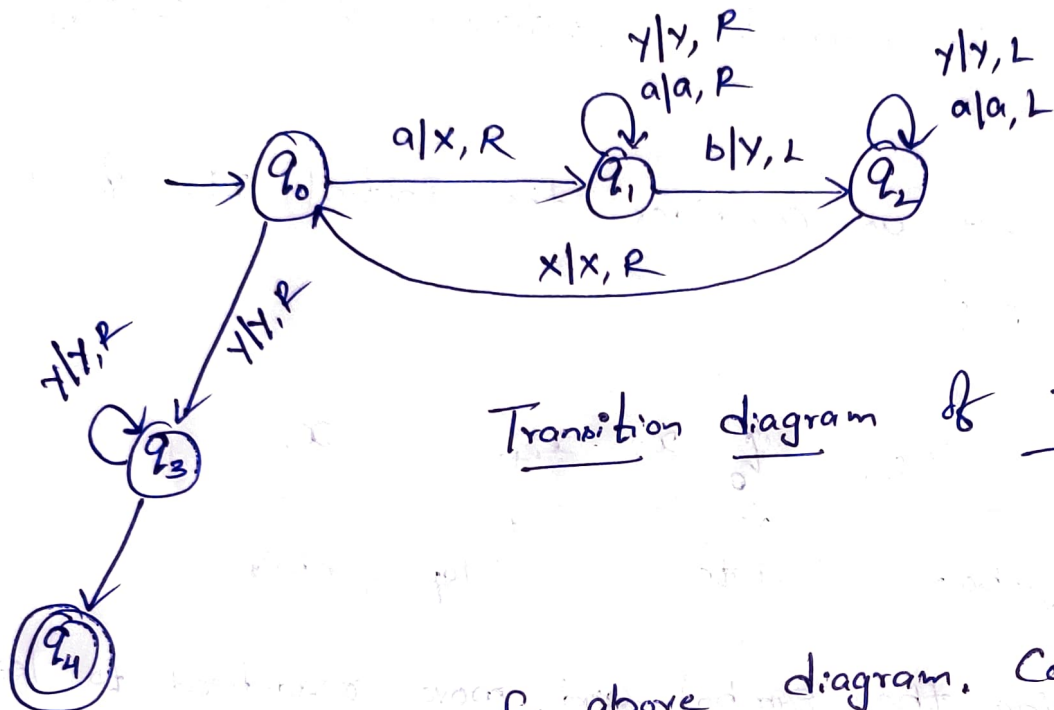
$$x_1, x_2, \dots, x_{i-1}, q_0, x_i, x_{i+1}, \dots, x_n \xrightarrow{\delta(q_0, x_i) \rightarrow (q_1, Y, R)} x_1, x_2, \dots, x_{i-1}, Y, q_1, x_{i+1}, \dots, x_n$$

Example : Create instantaneous description (ID)

for following language (L) :

Sol :

$$L = \{ a^n, b^n \mid n \geq 1 \}$$



Transition diagram of TM

- Now create ID for above diagram. Consider a string i.e 'aabb'

Initially  $q_0 aabb$

$q_0 aabb \vdash x q_1 abb \vdash$

$x q_1 bb \vdash x q_2 ayb \vdash$

$q_2 xayb \vdash x q_0 ayb \vdash$

$xx q_1 yb \vdash xx y q_1 b \vdash$

$xx q_2 yy \vdash x q_2 xyy \vdash$

$xx q_0 yy \vdash xx y q_3 y \vdash$

$xx yy q_3 \underline{\underline{B}} \vdash q_4$  Finally we reached to final state i.e string accepted by TM.

Note: on creating ID we used ' $\vdash$ ' symbol called "Turnstile" for moves on tape

# Decidability and Undecidability

## Recursive Language:

- A language 'L' is said to be recursive if there exists a Turing machine which will accept all the strings in 'L' and reject all strings not in 'L'.
- The Turing machine will halt every time and give an answer (accept or reject) for each and every string input.

## Recursively Enumerable Language:

- A Language 'L' is said to be recursively enumerable language if there exists a Turing machine which will accept (and therefore halt) for all the input strings which are in 'L'.
- But may or may not halt for all input strings which are not in 'L'.

## Decidable Language :

- A language 'L' is decidable if it is recursive language. All decidable languages are recursive languages & vice versa.

## Undecidable Language :

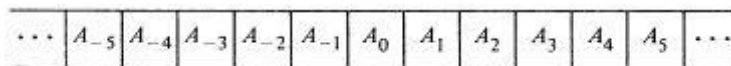
- A language is undecidable if it is not decidable.
- An undecidable language may sometimes be partially decidable but not decidable.
- If a language is not even partially decidable, then there exists no Turing machine for that language.



### **Types of Turing Machines:**

#### **i) Two-way infinite tape:**

A Turing machine with a two-way infinite tape is denoted by  $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ . As its name implies, the tape is infinite to the left as well as to the right. We denote an ID of such a device as for the one-way infinite TM. We imagine, however, that there is an infinity of blank cells both to the left and right of the current nonblank portion of the tape.

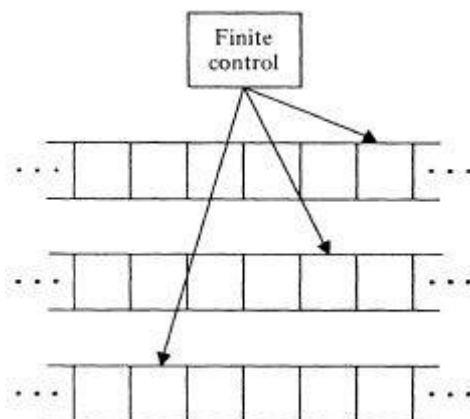


## ii) Multitape Turing machines:

A multitape Turing machine consists of a finite control with  $k$  tape heads and  $k$  tapes; each tape is infinite in both directions.. On a single move, depending on the state of the finite control and the symbol scanned by each of the tape heads, the machine can:

- 1) change state;
- 2) print a new symbol on each of the cells scanned by its tape heads;
- 3) move each of its tape heads,, independently, one cell to the left or right, or keep it stationary.

Initially, the input appears on the first tape, and the other tapes are blank.



## iii) Nondeterministic Turing machines:

A nondeterministic Turing machine is a device with a finite control and a single, one-way infinite tape. For a given state and tape symbol scanned by the tape head, the machine has a finite number of choices for the next move. Each choice consists of a new state, a tape symbol to print, and a direction of head motion. Note that the nondeterministic TM is not permitted to make a move in which the next state is selected from one choice, and the symbol printed and/or direction of head motion are selected from other choices. The nondeterministic TM accepts its input if any sequence of choices of moves leads to an accepting state.

**iv) Multidimensional Turing machines:**

The device has the usual finite control, but the tape consists of a  $k$ -dimensional array of cells infinite in all  $2k$  directions, for some fixed  $k$ . Depending on the state and symbol scanned, the device changes state, prints a new symbol, and moves its tape head in one of  $2k$  directions, either positively or negatively, along one of the  $k$  axes. Initially, the input is along one axis, and the head is at the left end of the input. At any time, only a finite number of rows in any dimension contain nonblank symbols, and

|       |          |          |          |       |          |          |
|-------|----------|----------|----------|-------|----------|----------|
| $B$   | $B$      | $B$      | $a_1$    | $B$   | $B$      | $B$      |
| $B$   | $B$      | $a_2$    | $a_3$    | $a_4$ | $a_5$    | $B$      |
| $a_6$ | $a_7$    | $a_8$    | $a_9$    | $B$   | $a_{10}$ | $B$      |
| $B$   | $a_{11}$ | $a_{12}$ | $a_{13}$ | $B$   | $a_{14}$ | $a_{15}$ |
| $B$   | $B$      | $a_{16}$ | $a_{17}$ | $B$   | $B$      | $B$      |

these rows each have only a finite number of nonblank symbols.

**v) Multihead Turing machines:**

A  $k$ -head Turing machine has some fixed number,  $k$ , of heads. The heads are numbered 1 through  $k$ , and a move of the TM depends on the state and on the symbol scanned by each head. In one move, the heads may each move independently left, right, or remain stationary.

**vi) Off-line Turing machines:**

An off-line Turing machine is a multitape TM whose input tape is read-only. Usually we surround the input by endmarkers,  $\varphi$  on the left and  $\$$  on the right. The Turing machine is not allowed to move the input tape head off the region between  $\varphi$  and  $\$$ .